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GreyCats Analytics — AI Features Release Notes

Version: 1.0.0 | **Date:** May 2026 | **Branch:** [feature/ai-suite](#)

Executive Summary

This release adds a complete AI-powered marketing intelligence suite to GreyCats Analytics, covering 4 core features as defined in the PRD: AI Report Chatbot, Train

Your Brand, AI Content Calendar, and AI Creative Suite — plus deep integrations with the existing Social Media Scheduler, Blog Scheduler, and Broadcast system.

Features Delivered

1. AI Report Chatbot

Location: Report Builder page (side panel)

- SSE streaming chat with smooth rAF-buffered rendering
- Context-aware: reads all report metrics, brand profile, content calendar, and scheduled posts
- Previous-period comparison (% change vs prior period)
- Monthly/weekly trend data for long date ranges
- Actionable suggestions: AI generates content actions inline per message
- "Add to Calendar" on action cards → flows to scheduler
- Follow-up question suggestions after each response
- Thumbs up/down feedback with DB persistence
- Conversation export (.txt download)
- Auto-generated session titles
- Rate limiting: 100 messages per user per report per day
- Brand voice applied to all responses

2. Train Your Brand

Location: Client Detail → Brand AI tab

- Website URL scraping with AI auto-fill (cheerio + AI extraction)
- SSRF protection (blocks private IPs, IPv6, file://, .local, .internal)
- 4-tab form: Identity / Voice & Style / Guidelines / Knowledge Base
- Fields: brandName, tagline, industry, brandVoice (6 visual options), targetAudience, valueProposition, missionStatement, brandStory, colorPalette, competitors, keyProducts, socialHandles, doList, dontList, additionalContext
- Knowledge base file upload: PDF (pdf-parse), DOCX (mammoth), TXT, PNG, JPG — text extracted and injected into all AI calls

- Brand context injected into ALL AI features (chatbot, calendar, creative, actions, suggestions, summary)

3. AI Content Calendar

Location: Backend API (used by chatbot actions + scheduler suggestions)

- Async calendar generation with progress polling (0-100%)
- AI generates full month of content: dates, platforms, content types, captions, hashtags, best times, data insights
- Historical performance data fed into generation (last 60 days metrics + published posts)
- CRUD: create, read, update, delete entries
- Entry edit dialog: context-aware generate button (Article / Video Script / Image)
- CSV export
- "From AI Chat" badge on chatbot-originated entries
- Approve/Reject/Undo workflow
- Convert to ScheduledPost with markdown → Unicode bold conversion
- Article teaser extraction for social media posts

4. AI Creative Suite

Location: Backend API (used by scheduler, blog, broadcasts)

- **Caption generation:** multi-caption with hashtags, platform char limits, brand voice
- **Article/script generation:** 500-800 word articles, 30-60s video scripts, markdown formatted
- **Image generation:** gpt-image-1 via OpenAI, async with polling, b64_json handling, saved to disk
- Asset gallery with folder/collection grouping by topic
- Rate limiting: 50 captions/day, 30 content/day, 20 images/day

5. Chatbot → Scheduler Bridge

Location: Social Media Scheduler page

- "Suggestions" button with badge count in scheduler header
- Content Suggestions modal: review, edit caption, set date/time, generate image, schedule
- Converts AI drafts to real ScheduledPost records
- Markdown → Unicode bold conversion for social media
- Article teaser for long-form content
- Image URL persistence across modal close/reopen

6. Write with AI (Inline)

Location: All text input fields across the platform

- **Social Media Post modal:** Sparkles icon in caption textarea + "Automate with AI" tab
- **Blog Post modal:** Sparkles in title input + body editor (Tiptap)
- **Broadcast Template Manager:** Sparkles in SMS/Email content textarea
- **Broadcast Create modal:** Sparkles in email subject input
- Behavior: empty → floating card asks topic; has text → AI rewrites/improves
- SMS: capped at 160 chars; Email: generates HTML with inline styles

Technical Architecture

New Backend Files

File	Purpose
controllers/aiChatController.ts	Chat sessions, streaming, actions, feedback, summary, analytics
controllers/brandController.ts	Brand profile CRUD, website scraping, knowledge base
controllers/calendarController.ts	Calendar generation, entries, scheduling

File	Purpose
<code>controllers/creativeController.ts</code>	Captions, content, images, assets
<code>controllers/aiAdminController.ts</code>	AI usage analytics
<code>services/ai/aiStreamService.ts</code>	SSE streaming for Anthropic, OpenAI, Google
<code>services/ai/reportContextBuilder.ts</code>	Report metric context with trends + comparison
<code>services/ai/brandContextService.ts</code>	Brand profile + knowledge base text builder
<code>services/ai/brandScraperService.ts</code>	Website scraping + AI brand extraction
<code>services/ai/historicalInsightsService.ts</code>	Historical metrics for content planning
<code>services/ai/aiProviderService.ts</code>	Multi-provider adapter (existing, enhanced)
<code>utils/sanitize.ts</code>	XSS prevention, AI output validation

New Frontend Files

File	Purpose
<code>components/ai/ReportChatbot.tsx</code>	Full chatbot UI with streaming
<code>components/clients/BrandSettings.tsx</code>	Brand AI settings form
<code>components/calendar/ContentCalendar.tsx</code>	Calendar generation + management
<code>components/creative/CreativeSuite.tsx</code>	Caption/image generation + gallery
<code>features/social-media/components/AISuggestionsPanel.tsx</code>	Scheduler suggestions modal

File	Purpose
hooks/useAIStream.ts	SSE streaming hook with rAF buffering
api/brandApi.ts	Brand API module
api/calendarApi.ts	Calendar API module
api/creativeApi.ts	Creative API module
api/chatApi.ts	Chat API module (enhanced)

New Prisma Models

Model	Fields
BrandProfile	websiteUrl, brandName, tagline, industry, brandVoice, targetAudience, valueProposition, missionStatement, brandStory, colorPalette, competitors, keyProducts, socialHandles, doList, dontList, additionalContext, logoUrl
BrandKnowledge	type, label, fileName, filePath, fileUrl, mimeType, fileSize, extractedText
AICalendar	month, year, platforms, goals, postsPerWeek, status, progress
AICalendarEntry	date, dayOfWeek, platform, contentType, postIdea, caption, hashtags, bestPostTime, dataInsight, imageUrl, source, status
AICreativeAsset	type, platform, goal, topic, captions, imageUrl, imagePath, promptUsed, aspectRatio, style
AIChatSession	templateId, clientId, dateFrom, dateTo, status, title, contextSnapshot
AIChatMessage	role, content, actions, rating

API Endpoints (27 total)

- **Brand:** 6 endpoints (`/api/ai/brand/*`)
- **Chat:** 9 endpoints (`/api/ai/chat/*`)
- **Calendar:** 7 endpoints (`/api/ai/calendar/*`)
- **Creative:** 5 endpoints (`/api/ai/creative/*`)

Security Measures

Measure	Implementation
SSRF Protection	Block private IPs, IPv6, file://, .local, .internal in brand scraper
XSS Prevention	<code>sanitizeText()</code> strips script/iframe/event handlers from all user + AI content
AI Output Validation	<code>sanitizeActions()</code> , <code>sanitizeCalendarEntries()</code> , <code>sanitizeBrandData()</code> — validate enums, cap lengths, strip HTML
Rate Limiting	Redis-based per-user-per-feature-per-day limits (fail-open on Redis down)
Auth	JWT Bearer token required on all endpoints, ownership verified on every mutation
File Upload	Multer with type whitelist (PDF/DOCX/TXT/PNG/JPG), 10MB max, PDF parse capped at 25MB
AI Timeout	120s max for any AI provider call via <code>withTimeout()</code> wrapper
SSE Headers	<code>X-Content-Type-Options: nosniff, charset=utf-8, no-store</code>
Image Filenames	<code>randomUUID()</code> instead of <code>Date.now()</code>
Race Conditions	Calendar creation uses try/catch fallback for concurrent requests

Testing

Unit Tests: 154 tests, 5 suites, ALL PASSING

Suite	Tests	Coverage
brandController.test.ts	26	Scrape, CRUD, knowledge base, SSRF
calendarController.test.ts	34	Generation, entries, scheduling, validation
creativeController.test.ts	31	Captions, content, images, assets
chatController.test.ts	29	Sessions, streaming, feedback, actions
services.test.ts	34	SSRF, brand context, markdown conversion

Build Status

Target	Status
Backend TypeScript (tsc --noEmit)	PASS — 0 errors
Frontend TypeScript (tsc --noEmit)	PASS — 0 errors
Frontend Production Build (vite build)	PASS — built in 11.78s

Manual Test Cases

- 187 test cases documented in requirements/AI_Features_Test_Cases.md
- All tested and verified

Configuration Required

Environment Variables


```
# AI Provider (system defaults)
AI_TEXT_PROVIDER=openai-compatible
AI_TEXT_MODEL=gpt-4o
AI_TEXT_BASE_URL=https://api.openai.com/v1
AI_IMAGE_PROVIDER=openai
AI_IMAGE_MODEL=gpt-image-1

# API Keys
OPENAI_API_KEY=sk-...
OPENAI_ORG_ID=org-...

# Existing (required)
APP_ENCRYPTION_KEY=...
REDIS_URL=redis://localhost:6379
```

NPM Dependencies Added

cheerio	– HTML parsing for website scraping
pdf-parse	– PDF text extraction
mammoth	– DOCX text extraction

Deliverables Checklist

- ☒ All 4 PRD features implemented (Chatbot, Calendar, Creative, Brand)
- ☒ Chatbot → Scheduler bridge with action cards
- ☒ Write with AI in Social Media, Blog, Broadcasts
- ☒ Brand context injected into all AI calls
- ☒ 27 API endpoints documented ([AI_API_Documentation_Complete.md](#))
- ☒ 187 test cases documented ([AI_Features_Test_Cases.md](#))
- ☒ 154 unit tests passing
- ☒ Security audit completed and fixes applied
- ☒ Zero TypeScript errors (backend + frontend)
- ☒ Production build successful